

EQ Credit: Acoustic Performance

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To provide workspaces and classrooms that promote occupants' well-being, productivity, and communications through effective acoustic design.

REQUIREMENTS

NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

For all occupied spaces, meet two of the following: HVAC background noise, Sound Transmission, and/or Reverberation time. Meet all three for an exemplary performance point.

Confirm compliance via calculations or measurements in representative rooms, and/or design documentation from a person experienced in the field of acoustics.

HVAC Background Noise

Achieve maximum background noise levels from heating, ventilating, and air conditioning (HVAC) systems per 2015 ASHRAE Handbook-- HVAC Applications, Chapter 48, Table 1 ; AHRI Standard 885-2008, Table 15; or a local equivalent.

If confirming compliance via measurements, use a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation, the International Electrotechnical Commission (2013) IEC 61672-1:2013 Electroacoustics – Sound Level Meters – Part 1: Specifications, or a local equivalent.

Comply with design criteria for HVAC noise levels resulting from the sound transmission paths listed in 2015 ASHRAE Handbook—HVAC Applications, Chapter 48, Table 6; or a local equivalent.

Sound Transmission

Categorize all occupied spaces by use and desired level of acoustic privacy.

Meet the composite sound transmission class (STC_c) ratings or noise isolation class (NIC) listed in Table 1. For NIC measurements, use ASTM E336-17a or Annex A.3 of ANSI S12.60-2010.

Table 1. Minimum composite sound transmission class ratings or noise isolation class for adjacent spaces

Adjacency combinations		STC _c **	NIC**
Retail	Retail	50	45
Collaborative / multi-use	Hallway, stairway	25	20
Private	Hallway, stairway	35	30
Confidential	Hallway, stairway	40	35
Collaborative / multi-use	Collaborative / multi-use	35	30
Collaborative / multi-use	Private	45	40
Collaborative / multi-use	Confidential	50	45
Private	Private	45	40

Private	Confidential	50	45
Confidential	Confidential	50	45
Conference room	Conference room	50	45
Mechanical equipment room*	Hallway, stairway	50	45
Mechanical equipment room*	Occupied area	60	55

*Minimum STCc or NIC has to be met unless proven that the equipment noise in conjunction with the sound isolation performance of the partitions and doors will not exceed the maximum background noise requirements of the adjacent space.

**If a sound masking system is implemented at a minimum level of 40 dBA, the STCc ratings or NIC values in Table 1 may be lowered by 5 points. This applies to all space types except mechanical equipment rooms. The sound masking system must be designed by an acoustical professional and meet the following criteria:

- ▶ The overall level for sound masking must be set by an acoustical professional and must not exceed 48 dBA in open offices, libraries, cafeterias, corridors/hallways, 45 dBA in enclosed offices, and 42 dBA in conference rooms, and wellness rooms. The combined level of masking and HVAC background noise must not exceed these limits.
- ▶ The system design and commissioning must provide overall level uniformity of +/-1 dBA and one-third octave band uniformity of +/-2 dB from at least 100 to 5,000 Hz when tested according to ASTM E1573-18
- ▶ The sound masking spectrum must conform to the National Research Council of Canada COPE Optimum Masking Spectrum or an alternate spectrum if specified by an acoustical engineer.

Reverberation Time

Meet the reverberation time requirements in Table 2 (adapted from Table 9.1 in the Performance Measurement Protocols for Commercial Buildings³).

Table 2. Reverberation time requirements

Room type	Application	T60 (sec), at 500 Hz, 1000 Hz, and 2000 Hz
Hotel/motel	Individual room or suite	< 0.6
	Meeting or banquet room	< 0.8
Office building	Executive or private office	< 0.6
	Conference room	< 0.6
	Teleconference room	< 0.6
	Open-plan office without sound masking	< 0.8
	Open-plan office with sound masking	0.8
Courtroom	Unamplified speech	< 0.7
	Amplified speech	< 1.0
Performing arts space	Drama theaters, concert and recital halls	Varies by application
Laboratories	Testing or research with minimal speech communication	< 1.0
	Extensive phone use and speech communication	< 0.6
Church, mosque, synagogue	General assembly with critical music program	Varies by application
Library		< 1.0
Indoor stadium, gymnasium	Gymnasium and natatorium	< 2.0
	Large-capacity space with speech amplification	< 1.5

³ Adapted from ASHRAE (2007d), ASA (2008), ANSI (2002), and CEN (2007)

SCHOOLS

HVAC Background noise

Achieve a background noise level of 35 dBA or less from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces.

Sound Transmission

Design classrooms and other core learning spaces to meet the sound transmission class (STC) requirements of ANSI S12.60–2010 Part 1, or a local equivalent. Exterior windows must have an STC rating of at least 35, unless outdoor and indoor noise levels can be verified to justify a lower rating.

HEALTHCARE

Design the facility to meet Option 1 (1 point) and/or Option 2 (1 point).

Option 1. Speech Privacy, Sound Isolation, and Background Noise (1 point)

Speech Privacy and Sound Isolation

Design sound isolation to achieve speech privacy, acoustical comfort, and minimal annoyance from noise-producing sources. Consider sound levels at both source and receiver locations, the background sound at receiver locations, and the occupants' acoustical privacy and acoustical comfort needs.

Design the facility to meet the criteria outlined in the following sections, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-6.1.5 and Section 1.2-6.1.6
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.6 and Section 1.2-5.1.6.2
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.6

Background Noise

Consider background noise levels generated by all building mechanical-electrical-plumbing systems, air distribution systems and other facility noise sources under the purview of the project building design-construction team.

Design the facility to meet the criteria outlined in the following sections, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-6.1.4 (Table 1.2-5)
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.4 (Table 1.2-5)
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.4 (Table 2.5-5)

Calculate or measure sound levels in representative rooms and spaces of each type to confirm compliance with criteria in the above-referenced tables using a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation. For spaces not listed in Table 1.2-2, refer to ASHRAE 2015 Handbook, Chapter 48, Sound and Vibration Control, Table 1.

Option 2. Acoustical Finishes and Site Exterior Noise (1 point)

Meet the requirements for acoustical finishes and site exterior noise.

Acoustical Finishes

Specify materials, products systems installation details, and other design features to meet the following:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-5.1.3 (Table 1.2-4)
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.3 (Table 1.2-4)

- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.3 (Table 2.5-4)

Calculate or measure the average sound absorption coefficients for representative unoccupied rooms of each type in the building to confirm conformance with the requirements.

Site Exterior Noise

Minimize the effect on building occupants of site exterior noise produced by road traffic, aircraft flyovers, railroads, on-site heliports, emergency power generators during maintenance testing, outdoor facility MEP and building services equipment, etc. Also minimize effects on the surrounding community from all facility MEP equipment and activities as required to meet (1) local applicable codes or (2) background noise requirements above, whichever is more stringent.

Comply with the 2018 FGI Guidelines for the following noise sources:

- ▶ heliports, A1.3-3.6.2.2;
- ▶ generators, 2.1-8.3.3.1;
- ▶ mechanical equipment, 2.1-8.2.1.1; and
- ▶ building services, A2.2-5.3

Measure and analyze data to determine the exterior noise classification (A, B, C, or D) of the facility site and design the building envelope to meet the following, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Table 1.2-3
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities— Table 1.2-3
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities—Table 2.5-3

For exterior site exposure categories B, C, or D, calculate or measure the sound isolation performance of representative elements of the exterior building envelope to determine the composite sound transmission class (STCc) rating for representative façade sections. Measurements should generally conform to ASTM E966-18, Standard Guide for Field Measurements of Airborne Sound Insulation of Building Façades and Façade Elements.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The credit now requires any two of three sound performance requirements to be met in form of HVAC background noise, sound transmission/isolation and reverberation time to provide more flexibility to project teams. Further, projects can comply with an International Electrotechnical Commission standard for HVAC background noise measurement. Minimum sound transmission class ratings table is significantly revised to include multiple space types and adjacency combinations Projects can now meet either minimum STC ratings or corresponding minimum NIC ratings

Sound reinforcement requirements have been removed and sound masking requirements are modified into sound transmission/isolation with clearer criteria for sound masking in different space types concurrent with any HVAC background noise in the spaces.

Finally, some existing standards referenced for this credit were changed to their most recent version as follows:

- ▶ 2011 ASHRAE Handbook- HVAC Applications → 2015 ASHRAE Handbook- HVAC Applications
- ▶ Healthcare: FGI 2010 → FGI 2018

Step-by-Step Guidance

Follow LEED v4 reference guide with the following modifications:

Step 2: Review Acoustic Criteria

Meet any two of three performance areas from HVAC background noise, sound isolation (that includes sound masking system requirements) and reverberation time instead of all four.

Step 3: Address HVAC background noise. Follow LEED v4 reference guide for this section with the exception of referring to newer 2015 HVAC Applications ASHRAE Handbook.

Step 4: Verify HVAC background noise. Follow LEED v4 reference guide for this section in addition to referring to newly introduced standard for HVAC background noise measurement –International Electrotechnical Commission IEC 61672-1:2013 Electroacoustics – Sound Level Meters- Part 1 Specifications or a local equivalent.

Step 7: Verify sound isolation. Follow LEED v4 reference guide for this section. A NIC rating within 3-5 points of the specified STC rating may be considered compliant. For each space, list the maximum STC or NIC rating, design STC or NIC rating and data to support reported values. Determine whether the project will have any associated sound masking systems.

Step 11: Select sound reinforcement and masking systems. Follow the credit requirements under sound transmission/isolation for sound level and system uniformity for the specific spaces using sound masking systems.

Healthcare

Step 2: Follow the LEED v4 reference guide with the exception of referring to 2015 FGI Guidelines for Design and Construction of Health Care Facilities and the Sound and Vibration Design Guidelines for Health Care Facilities.

Further Explanation

Follow the LEED v4 Reference Guide with the following modifications:

Masking systems and meeting masking system requirements

Follow LEED v4 reference guide for these sections. Comply with newer requirements in revised credit for sound masking- meet STC or NIC ratings as well as HVAC background noise levels in conjunction with sound masking requirements for specific space types having varied environmental noise levels (as specified for open spaces, enclosed offices or conference rooms etc.)

Required Documentation

Follow the LEED v4 reference guide with the following modifications:

- ▶ For sound isolation requirements of all projects, either STC or NIC ratings for space adjacencies are acceptable.
- ▶ Sound masking systems documentation needs to be submitted under sound isolation rather than separately.
- ▶ Healthcare: For site exterior noise (Option 2), provide mitigation narrative for each 2015 FGI guideline.

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to minimize HVAC background noise, exterior noise, reverberation time etc. in newly constructed buildings can help contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.